Mahi Prashant Nakhate

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In [ ]:
```
import pandas as pd
import numpy as np
import seaborn as sns

import matplotlib.pyplot as plt
%matplotlib inline

import warnings
warnings.filterwarnings("ignore")
```

In [10]:
```
df = pd.read_csv('voice.csv')
df.head()
```

Out[10]:

	meanfreq	sd	median	Q25	Q75	IQR	skew	kurt	sp.ent	sfm	...	centroid	meanfun	minfun	max
0	0.059781	0.064241	0.032027	0.015071	0.090193	0.075122	12.863462	274.402906	0.893369	0.491918	...	0.059781	0.084279	0.015702	0.275
1	0.066009	0.067310	0.040229	0.019414	0.092666	0.073252	22.423285	634.613855	0.892193	0.513724	...	0.066009	0.107937	0.015826	0.250
2	0.077316	0.083829	0.036718	0.008701	0.131908	0.123207	30.757155	1024.927705	0.846389	0.478905	...	0.077316	0.098706	0.015656	0.271
3	0.151228	0.072111	0.158011	0.096582	0.207955	0.111374	1.232831	4.177296	0.963322	0.727232	...	0.151228	0.088965	0.017798	0.250
4	0.135120	0.079146	0.124656	0.078720	0.206045	0.127325	1.101174	4.333713	0.971955	0.783568	...	0.135120	0.106398	0.016931	0.266

5 rows × 21 columns

In [11]:
```
df.shape
```

Out[11]:
```
(3168, 21)
```

In [12]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3168 entries, 0 to 3167
Data columns (total 21 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   meanfreq  3168 non-null   float64
 1   sd        3168 non-null   float64
 2   median    3168 non-null   float64
 3   Q25       3168 non-null   float64
 4   Q75       3168 non-null   float64
 5   IQR       3168 non-null   float64
 6   skew      3168 non-null   float64
 7   kurt      3168 non-null   float64
 8   sp.ent    3168 non-null   float64
 9   sfm       3168 non-null   float64
 10  mode      3168 non-null   float64
 11  centroid  3168 non-null   float64
 12  meanfun   3168 non-null   float64
 13  minfun    3168 non-null   float64
 14  maxfun    3168 non-null   float64
 15  meandom   3168 non-null   float64
 16  mindom    3168 non-null   float64
 17  maxdom    3168 non-null   float64
 18  dfrange   3168 non-null   float64
 19  modindx   3168 non-null   float64
 20  label     3168 non-null   object 
dtypes: float64(20), object(1)
memory usage: 519.9+ KB
```

In [13]:
```
df.describe()
```

Out[13]:

	meanfreq	sd	median	Q25	Q75	IQR	skew	kurt	sp.ent	sfm	
count	3168.000000	3168.000000	3168.000000	3168.000000	3168.000000	3168.000000	3168.000000	3168.000000	3168.000000	3168.000000	3168.00
mean	0.180907	0.057126	0.185621	0.140456	0.224765	0.084309	3.140168	36.568461	0.895127	0.408216	0.16
std	0.029918	0.016652	0.036360	0.048680	0.023639	0.042783	4.240529	134.928661	0.044980	0.177521	0.07
min	0.039363	0.018363	0.010975	0.000229	0.042946	0.014558	0.141735	2.068455	0.738651	0.036876	0.00
25%	0.163662	0.041954	0.169593	0.111087	0.208747	0.042560	1.649569	5.669547	0.861811	0.258041	0.11
50%	0.184838	0.059155	0.190032	0.140286	0.225684	0.094280	2.197101	8.318463	0.901767	0.396335	0.18
75%	0.199146	0.067020	0.210618	0.175939	0.243660	0.114175	2.931694	13.648905	0.928713	0.533676	0.22
max	0.251124	0.115273	0.261224	0.247347	0.273469	0.252225	34.725453	1309.612887	0.981997	0.842936	0.28

In [14]:
```
df.corr()
```

Out[14]:

	meanfreq	sd	median	Q25	Q75	IQR	skew	kurt	sp.ent	sfm	mode	centroid	meanfu
meanfreq	1.000000	-0.739039	0.925445	0.911416	0.740997	-0.627605	-0.322327	-0.316036	-0.601203	-0.784332	0.687715	1.000000	0.46084
sd	-0.739039	1.000000	-0.562603	-0.846931	-0.161076	0.874660	0.314597	0.346241	0.716620	0.838086	-0.529150	-0.739039	-0.46628
median	0.925445	-0.562603	1.000000	0.774922	0.731849	-0.477352	-0.257407	-0.243382	-0.502005	-0.661690	0.677433	0.925445	0.41490
Q25	0.911416	-0.846931	0.774922	1.000000	0.477140	-0.874189	-0.319475	-0.350182	-0.648126	-0.766875	0.591277	0.911416	0.54503
Q75	0.740997	-0.161076	0.731849	0.477140	1.000000	0.009636	-0.206339	-0.148881	-0.174905	-0.378198	0.486857	0.740997	0.15509
IQR	-0.627605	0.874660	-0.477352	-0.874189	0.009636	1.000000	0.249497	0.316185	0.640813	0.663601	-0.403764	-0.627605	-0.53446
skew	-0.322327	0.314597	-0.257407	-0.319475	-0.206339	0.249497	1.000000	0.977020	-0.195459	0.079694	-0.434859	-0.322327	-0.16766
kurt	-0.316036	0.346241	-0.243382	-0.350182	-0.148881	0.316185	0.977020	1.000000	-0.127644	0.109884	-0.406722	-0.316036	-0.19456
sp.ent	-0.601203	0.716620	-0.502005	-0.648126	-0.174905	0.640813	-0.195459	-0.127644	1.000000	0.866411	-0.325298	-0.601203	-0.51319
sfm	-0.784332	0.838086	-0.661690	-0.766875	-0.378198	0.663601	0.079694	0.109884	0.866411	1.000000	-0.485913	-0.784332	-0.42106
mode	0.687715	-0.529150	0.677433	0.591277	0.486857	-0.403764	-0.434859	-0.406722	-0.325298	-0.485913	1.000000	0.687715	0.32477
centroid	1.000000	-0.739039	0.925445	0.911416	0.740997	-0.627605	-0.322327	-0.316036	-0.601203	-0.784332	0.687715	1.000000	0.46084
meanfun	0.460844	-0.466281	0.414909	0.545035	0.155091	-0.534462	-0.167668	-0.194560	-0.513194	-0.421066	0.324771	0.460844	1.00000
minfun	0.383937	-0.345609	0.337602	0.320994	0.258002	-0.222680	-0.216954	-0.203201	-0.305826	-0.362100	0.385467	0.383937	0.33938
maxfun	0.274004	-0.129662	0.251328	0.199841	0.285584	-0.069588	-0.080861	-0.045667	-0.120738	-0.192369	0.172329	0.274004	0.31195
meandom	0.536666	-0.482726	0.455943	0.467403	0.359181	-0.333362	-0.336848	-0.303234	-0.293562	-0.428442	0.491479	0.536666	0.27084
mindom	0.229261	-0.357667	0.191169	0.302255	-0.023750	-0.357037	-0.061608	-0.103313	-0.294869	-0.289593	0.198150	0.229261	0.16216
maxdom	0.519528	-0.482278	0.438919	0.459683	0.335114	-0.337877	-0.305651	-0.274500	-0.324253	-0.436649	0.477187	0.519528	0.27798
dfrange	0.515570	-0.475999	0.435621	0.454394	0.335648	-0.331563	-0.304640	-0.272729	-0.319054	-0.431580	0.473775	0.515570	0.27515
modindx	-0.216979	0.122660	-0.213298	-0.141377	-0.216475	0.041252	-0.169325	-0.205539	0.198074	0.211477	-0.182344	-0.216979	-0.05485

In [15]:
```
df.isnull().sum()
```

Out[15]:
```
meanfreq    0
sd          0
median      0
Q25         0
Q75         0
IQR         0
skew        0
kurt        0
sp.ent      0
sfm         0
mode        0
centroid    0
meanfun     0
minfun      0
maxfun      0
meandom     0
mindom      0
maxdom      0
dfrange     0
modindx     0
label       0
dtype: int64
```

In [17]:
```
print("Total number of labela: {}".format(df.shape[0]))
print("Number of male: {}".format(df[df.label == 'male'].shape[0]))
print("Number of female: {}".format(df[df.label == 'female'].shape[0]))
```

```
Total number of labela: 3168
Number of male: 1584
Number of female: 1584
```

In [18]:
```
X = df.iloc[:, :-1]
X.head()
```

Out[18]:

	meanfreq	sd	median	Q25	Q75	IQR	skew	kurt	sp.ent	sfm	mode	centroid	meanfun	minfun
0	0.059781	0.064241	0.032027	0.015071	0.090193	0.075122	12.863462	274.402906	0.893369	0.491918	0.000000	0.059781	0.084279	0.015702
1	0.066009	0.067310	0.040229	0.019414	0.092666	0.073252	22.423285	634.613855	0.892193	0.513724	0.000000	0.066009	0.107937	0.015826
2	0.077316	0.083829	0.036718	0.008701	0.131908	0.123207	30.757155	1024.927705	0.846389	0.478905	0.000000	0.077316	0.098706	0.015656
3	0.151228	0.072111	0.158011	0.096582	0.207955	0.111374	1.232831	4.177296	0.963322	0.727232	0.083878	0.151228	0.088965	0.017798
4	0.135120	0.079146	0.124656	0.078720	0.206045	0.127325	1.101174	4.333713	0.971955	0.783568	0.104261	0.135120	0.106398	0.016931

In [22]:
```
from sklearn.preprocessing import LabelEncoder
y = df.iloc[:, -1]

# Encode label category
# male -> 1
# female -> 0

gender_encoder = LabelEncoder()
y = gender_encoder.fit_transform(y)
y
```

Out[22]:
```
array([1, 1, 1, ..., 0, 0, 0])
```

In [23]:
```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
scaler.fit(X)
X = scaler.transform(X)
```

In [30]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 1)
```

In [32]:
```
from sklearn.svm import SVC
from sklearn import metrics
svc = SVC()     #Default hyperparameters
svc.fit(X_train, y_train)
y_pred = svc.predict(X_test)
print("Accuracy Score:")
print(metrics.accuracy_score(y_test, y_pred))
```

```
Accuracy Score:
0.9763406940063092
```

Different Hyperplane in SVC

In [39]:
```
svc = SVC(kernel = 'linear')
svc.fit(X_train,y_train)
y_pred = svc.predict(X_test)
print('Accuarcy Score:')
print(metrics.accuracy_score(y_test, y_pred))
```

```
Accuarcy Score:
0.9779179810725552
```

In [40]:
```
svc = SVC(kernel = 'rbf')
svc.fit(X_train,y_train)
y_pred = svc.predict(X_test)
print('Accuarcy Score:')
print(metrics.accuracy_score(y_test, y_pred))
```

```
Accuarcy Score:
0.9763406940063092
```

In [41]:
```
svc = SVC(kernel = 'poly')
svc.fit(X_train,y_train)
y_pred = svc.predict(X_test)
print('Accuarcy Score:')
print(metrics.accuracy_score(y_test, y_pred))
```

```
Accuarcy Score:
0.9589905362776026
```

In [42]:
```
svc = SVC(kernel = 'sigmoid')
svc.fit(X_train,y_train)
y_pred = svc.predict(X_test)
print('Accuarcy Score:')
print(metrics.accuracy_score(y_test, y_pred))
```

```
Accuarcy Score:
0.7933753943217665
```

In [ ]:

In [ ]:

In [ ]:

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In [ ]:

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